IN THE QUASI-COHERENT SITUATION OF A RINGED SITE THERE IS A CARDINAL KAPPA BOUNDING THE SIZE OF A NONZERO MAP TO ANY NONZERO OBJECT AND OF A FACTORISATION THROUGH A DIRECT SUM

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is the author's complete argument: the three-cardinal definition of kappa, the reduction of both assertions to the claim, and the induction proving the claim with the choice of the subcomplexes N_f, M_f and the application of lemma-build-from are all written out, ending with the colimit argument that H represents an object of QC(O).

Editorial note. The label lemma-cartesian-colimit-kappa names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

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Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The category QC(O) of quasi-coherent objects of D(O). Same chapter, the full subcategory QC(O) of D(O) defined by the condition that the derived tensor comparison maps are isomorphisms and its presentation by complexes. Supplies the ambient category in which both properties are stated and in which the complexes of size <= kappa are chosen.
- Size |K| of an object of QC(O). Same chapter, the cardinality |K| of an object of QC(O) defined as the minimum size of a complex of O-modules representing it. Supplies the quantity bounded by kappa in both assertions of the statement and in the inductive construction.
- The quasi-compact, quasi-separated local situation of the section. Same chapter, the standing situation above the lemma (the site C with its structural sheaf O and the auxiliary objects j_{U!}O_U). Supplies the coproducts appearing in the definition of kappa and the objects used in part (1).

Source reference. Original lemma 0GYX: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 11217–11217. Prerequisite pointer invoked at proof line(s) 11434; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 0GYY: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 11238–11238. Prerequisite pointer invoked at proof line(s) 11318; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0GYI: sources/stacks-chow/tags.txt, source lines 21143–21143. Prerequisite pointer invoked at proof line(s) 11316; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In the situation above, there exists a cardinal κ with the following properties:*

1. *for every nonzero object K of $QC(\mathcal{O})$ there exists a nonzero morphism $E \to K$ of $QC(\mathcal{O})$ such that $|E| \leq \kappa$,*
2. *for every morphism $\alpha : E \to \bigoplus_n K_n$ of $QC(\mathcal{O})$ such that $|E| \leq \kappa$, there exist morphisms $E_n \to K_n$ in $QC(\mathcal{O})$ with $|E_n| \leq \kappa$ such that α factors through $\bigoplus E_n \to \bigoplus K_n$.*

Proof. Let κ be an upper bound for the following set of cardinals:

1. $|\coprod_V j_{U!}\mathcal{O}_U(V)|$ for all $U \in \mathrm{Ob}(\mathcal{C})$,
2. the cardinals $\kappa(\mathcal{O}(V) \to \mathcal{O}(U))$ found in More on Algebra, Lemma 0GYI for all morphisms $U \to V$ in $\mathcal{C}$,
3. the cardinal found in Lemma 0GYY.

We claim that for any complex $\mathcal{F}^\bullet$ representing an object of $QC(\mathcal{O})$ and any subcomplex $\mathcal{S}^\bullet \subset \mathcal{F}^\bullet$ with $|\mathcal{S}^\bullet| \leq \kappa$ there exists a subcomplex $\mathcal{H}^\bullet$ of $\mathcal{F}^\bullet$ containing $\mathcal{S}^\bullet$ such that $\mathcal{H}^\bullet$ represents an object of $QC(\mathcal{O})$ and such that $|\mathcal{H}^\bullet| \leq \kappa$. In the next two paragraphs we show that the claim implies the lemma.

As in (1) let K be a nonzero object of $QC(\mathcal{O})$. Say K is represented by the complex of $\mathcal{O}$-modules $\mathcal{F}^\bullet$. Then $H^i(\mathcal{F}^\bullet)$ is nonzero for some i. Hence there exists an object U of $\mathcal{C}$ and a section $s \in \mathcal{F}^i(U)$ with $d(s) = 0$ which determines a nonzero section of $H^i(\mathcal{F}^\bullet)$ over U. Then the image of $s : j_{U!}\mathcal{O}_U[-i] \to \mathcal{F}^\bullet$ is a subcomplex $\mathcal{S}^\bullet \subset \mathcal{F}^\bullet$ with $|\mathcal{S}^\bullet| \leq \kappa$. Applying the claim we get $\mathcal{H}^\bullet \to \mathcal{F}^\bullet$ in $QC(\mathcal{O})$ nonzero with $|\mathcal{H}^\bullet| \leq \kappa$. Thus (1) holds.

Let $\alpha : E \to \bigoplus K_n$ be as in (2). Choose any complexes $\mathcal{K}_n^\bullet$ representing K_n. Then $\bigoplus \mathcal{K}_n^\bullet$ represents $\bigoplus K_n$. By the construction of the derived category we can represent E by a complex $\mathcal{E}^\bullet$ such that α is represented by a morphism $a : \mathcal{E}^\bullet \to \bigoplus \mathcal{K}_n^\bullet$ of complexes. By Lemma 0GYY and our choice of κ above we may assume $|\mathcal{E}^\bullet| \leq \kappa$. By the claim we get subcomplexes $\mathcal{E}_n^\bullet \subset \mathcal{K}_n^\bullet$ representing objects E_n of $QC(\mathcal{O})$ with $|E_n| \leq \kappa$ containing the image of $a_n : \mathcal{E}^\bullet \to \mathcal{K}_n^\bullet$ as desired.

Proof of the claim. Let $\mathcal{F}^\bullet$ be a complex representing an object of $QC(\mathcal{O})$ and let $\mathcal{S}^\bullet \subset \mathcal{F}^\bullet$ be a subcomplex of size $\leq \kappa$. We are going to inductively construct subcomplexes

$$\mathcal{S}^\bullet = \mathcal{S}_0^\bullet \subset \mathcal{S}_1^\bullet \subset \mathcal{S}_2^\bullet \subset \ldots \subset \mathcal{F}^\bullet$$

of size $\leq \kappa$ such that for every morphism $f : U \to V$ of $\mathcal{C}$ and every $i \in \mathbf{Z}$

1. the kernel of the arrow $H^i(\mathcal{S}_n^\bullet(V) \otimes^{\mathbf{L}}_{\mathcal{O}(V)} \mathcal{O}(U)) \to H^i(\mathcal{S}_n^\bullet(U))$ maps to zero in $H^i(\mathcal{S}_{n+1}^\bullet(V) \otimes^{\mathbf{L}}_{\mathcal{O}(V)} \mathcal{O}(U))$,
2. the image of the arrow $H^i(\mathcal{S}_n^\bullet(U)) \to H^i(\mathcal{S}_{n+1}^\bullet(U))$ is contained in the image of $H^i(\mathcal{S}_{n+1}^\bullet(V) \otimes^{\mathbf{L}}_{\mathcal{O}(V)} \mathcal{O}(U)) \to H^i(\mathcal{S}_{n+1}^\bullet(U))$,

Once this is done we can set $\mathcal{H}^\bullet = \bigcup \mathcal{S}^\bullet_n$. Namely, since derived tensor product and taking cohomology of complexes of modules over rings commute with filtered colimits, the conditions (1) and (2) together will guarantee that

$$\mathcal{H}^\bullet(V) \otimes^{\mathbf{L}}_{\mathcal{O}(V)} \mathcal{O}(U) \longrightarrow \mathcal{H}^\bullet(U)$$

is an isomorphism on cohomology in all degrees and hence an isomorphism in $D(\mathcal{O}(U))$ for all $f : U \to V$ in $\mathcal{C}$. Hence $\mathcal{H}^\bullet$ represents an object of $QC(\mathcal{O})$ as desired.

Construction of $\mathcal{S}_{n+1}$ given $\mathcal{S}_n$. For every morphism $f : U \to V$ of $\mathcal{C}$ we consider the commutative diagram

$$\begin{array}{ccc} \mathcal{S}^\bullet_n(V) & \longrightarrow & \mathcal{S}^\bullet_n(U) \\ \downarrow & & \downarrow \\ \mathcal{F}^\bullet(V) & \longrightarrow & \mathcal{F}^\bullet(U) \end{array}$$

This is a diagram as in More on Algebra, Lemma 0GYI for the ring map $\mathcal{O}(V) \to \mathcal{O}(U)$, i.e., the bottom row induces an isomorphism

$$\mathcal{F}^\bullet(V) \otimes^{\mathbf{L}}_{\mathcal{O}(V)} \mathcal{O}(U) \longrightarrow \mathcal{F}^\bullet(U)$$

in $D(\mathcal{O}(U))$. Thus we may choose subcomplexes

$$\mathcal{S}^\bullet_n(V) \subset M^\bullet_f \subset \mathcal{F}^\bullet(V) \quad \text{and} \quad \mathcal{S}^\bullet_n(U) \subset N^\bullet_f \subset \mathcal{F}^\bullet(U)$$

as in More on Algebra, Lemma 0GYI and in particular we see that $|N^i_f|, |M^i_f| \leq \kappa$. Next, we apply Lemma 0GYX using the subsets

$$\mathcal{S}^i_n(U) \amalg \coprod\nolimits_{f:U\to V} N^i_f \amalg \coprod\nolimits_{g:W\to U} M^i_g \subset \mathcal{F}^i(U)$$

to find a subcomplex

$$\mathcal{S}^\bullet_n \subset \mathcal{S}^\bullet_{n+1} \subset \mathcal{F}^\bullet$$

with containing those subsets and such that $|\mathcal{S}^\bullet_{n+1}| \leq \kappa$. Conditions (1) and (2) hold because the corresponding statements hold for $\mathcal{S}^\bullet_n(V) \subset M^\bullet_f$ and $\mathcal{S}^\bullet_n(U) \subset N^\bullet_f$ by the construction in More on Algebra, Lemma 0GYI. Thus the proof is complete. $\square$